

Artificial Neural Network Detection of Pancreatic Cancer from Proton (1H) Magnetic Resonance Spectroscopy Patterns of Plasma Metabolites

Corresponding Author: Mr Meiyappan Solaiyappan

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

This paper focus on the application of artificial neural networks to plasma metabolite patterns using 1H magnetic resonance spectroscopy to detect pancreatic ductal adenocarcinoma. They used two different spectra, a CPMG pulse sequence and a single pulse ZGPR sequence. The main results were interesting since they found that the sensitivity in the blinded study of PDAC classification was 87.5% and the specificity was 93.1%. The AUC ROC was 0.931 and Precision-Recall curves 0.854. The spectral patterns that played a role in discriminating patients with PDAC from subjects without cancer were associated with lipids, glucose, lactate, acetate, glutamine, myoinositol, BHB and BCAA regions. These results need to be validated in larger cohorts of patients with early stage 1 and 2 PDAC and healthy subjects and patients with risk of pancreas cancer.

The paper is scientifically accurate and the authors report results in a critical and effective way. The results achieved are considerable given the limited data in the literature. The methods and the statistical analysis are well described. The figures and tables are informative. The Discussion is short but sufficient.

Minor comments

- The study included a training study with 56 PDAC patients (stage 1 (n=6), stage 2 (n=16), stage 3 (n=24), stage 4 (n=10)), 50 healthy subjects (some with diabetes) and 54 patients with non-malignant pancreas disease. The blinded test study included 14 patients with PDAC (stage 1 (n=1), stage 2 (n=8), stage 3 (n=3), stage 4 (n=2)), 19 healthy subjects (some with diabetes) and 10 patients with non-malignant pancreas disease. It is stated that plasma samples from 16 of the PDAC patients were collected before chemotherapy. When was the samples from the other PDAC patients collected? Before operation? During chemotherapy?
- Please correct the numbers in the text so it corresponds to the numbers given in Supplementary Figure 1. Please also correct the numbers in Supplementary Table 1 and Supplementary Figure 1. Why were outliers excluded in this figure?
- Some of the healthy subjects (number ?) had diabetes. Did any of the healthy subjects have high BMI? Did any of the PDAC patients have diabetes, cachexia or overweight? These factors could affect the plasma metabolite pattern.
- In the discussion the authors should comment on the limitation that very few PDAC patients in their cohort had low stage 1 and 2 disease. This is the group of patients that needs to be detected in a screening test for PDAC.
- Please change reference 5 to a more recent paper.

Reviewer #2

(Remarks to the Author)

Solaiyappan et al. presented an excellent work utilizing a neural network for PDAC Dx of 1H MR spectral patterns of plasma metabolites. The study included PDAC, benign disease, and controls and conducted a multi-class classification, an adequate design for translational biomarker research. A few comments are listed below for further consideration:

- Were all PDAC patients in the early stages? If not, a sub-group analysis that only accounts for early/very early stage PDAC should be tested.

- The introduction can be improved. If I am not mistaken, there have been quite a few reviews and systematic reviews on this issue. These references could be extremely helpful in illustrating the current issues of the field, emphasizing the need for additional studies. Besides, please consider removing the discussion related to proteomics or making the discussion more mature and meaningful.

- Studies using spectroscopy usually aim to extract meaningful signals with assigned metabolites. This study, instead, focused on patterns. I guess it would be nice to discuss the pros/cons/trade-offs.

- (Line 425-436) How are we certain that the neural-net selected the patterns in meaningful areas? Can we do anything to at least provide or speculate on the nature (e.g., possible classes of compounds) of the selected regions? Can we know which regions are of more importance than the rest and where misclassification may occur?

- The raw data and the analytical scripts should be publicly available, as it is generally a requirement for this category of studies.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The authors have addressed the earlier comments sufficient and thereby improved the manuscript sufficiently. I have no further comments to the manuscript.

Reviewer #2

(Remarks to the Author)

The authors did an excellent job in revising the manuscript. I can see that the revised manuscript has significantly improved in terms of clarity and data presentation.

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Reviewers' comments:

Reviewer #1 (Remarks to the Author):

This paper focuses on the application of artificial neural networks to plasma metabolite patterns using ¹H magnetic resonance spectroscopy to detect pancreatic ductal adenocarcinoma. They used two different spectra, a CPMG pulse sequence and a single pulse ZGPR sequence. The main results were interesting since they found that the sensitivity in the blinded study of PDAC classification was 87.5% and the specificity was 93.1%. The AUC ROC was 0.931 and Precision-Recall curves 0.854. The spectral patterns that played a role in discriminating patients with PDAC from subjects without cancer were associated with lipids, glucose, lactate, acetate, glutamine, myoinositol, BHB and BCAA regions. These results need to be validated in larger cohorts of patients with early stage 1 and 2 PDAC and healthy subjects and patients with risk of pancreas cancer.

The paper is scientifically accurate and the authors report results in a critical and effective way. The results achieved are considerable given the limited data in the literature. The methods and the statistical analysis are well described. The figures and tables are informative. The Discussion is short but sufficient.

Response: We thank the reviewer for these comments.

Minor comments

• *The study included a training study with 56 PDAC patients (stage 1 (n=6), stage 2 (n=16), stage 3 (n=24), stage 4 (n=10)), 50 healthy subjects (some with diabetes) and 54 patients with non-malignant pancreas disease. The blinded test study included 14 patients with PDAC (stage 1 (n=1), stage 2 (n=8), stage 3 (n=3), stage 4 (n=2)), 19 healthy subjects (some with diabetes) and 10 patients with non-malignant pancreas disease. It is stated that plasma samples from 16 of the PDAC patients were collected before chemotherapy. When was the samples from the other PDAC patients collected? Before operation? During chemotherapy?*

Response: We have provided this information in the revised manuscript. Briefly, in the PDAC training group plasma samples from 25 PDAC participants were obtained after neoadjuvant chemotherapy. Plasma samples from the remaining 32 PDAC participants were collected prior to chemotherapy. In the blinded data set, plasma samples from 7 patients were obtained after neoadjuvant chemotherapy and from 8 patients prior to chemotherapy. All plasma samples from patients with PDAC who underwent surgical resection were obtained prior to surgery. [Please see page 7, lines 267 - 273.](#)

• *Please correct the numbers in the text so it corresponds to the numbers given in Supplementary Figure 1. Please also correct the numbers in Supplementary Table 1 and Supplementary Figure 1. Why were outliers excluded in this figure?*

Response: Thank you for identifying this discrepancy. We have corrected the numbers to match the text in the revised manuscript. We have included the outliers in calculating the median in Supplementary Figure 1. In the revised figure, the single outliers in the different groups can be identified as a separate point. We have included this in the corresponding Figure Legend.

• *Some of the healthy subjects (number ?) had diabetes. Did any of the healthy subjects have high BMI? Did any of the PDAC patients have diabetes, cachexia or overweight? These factors could affect the plasma metabolite pattern.*

Response: The reviewer makes an excellent point. We have provided the distribution of high BMI, diabetes and cachexia in Supplementary Tables 1D-E, and included the details in the revised manuscript. [Please see page 7, lines 267 - 273, and page 13, lines 513 – 518.](#)

Briefly, the PDAC training group included participants with obesity (n=12), DM (n=28), and cachexia (n=20). The normal and disease training groups included participants with obesity (n=22, normal; n=14, disease) and diabetes mellitus (n=9, normal; n=10, disease). In the blinded data sets, healthy subjects,

and subjects in the disease and PDAC groups had high BMI (n=3, normal group; n=3, disease group; n=4, PDAC group) and DM (n=2, normal group; n=2, disease group; n=10, PDAC group). Approximately half the blinded PDAC subjects also had cachexia (n=8). The neural network was trained and tested within the background of these conditions.

• *In the discussion the authors should comment on the limitation that very few PDAC patients in their cohort had low stage 1 and 2 disease. This is the group of patients that needs to be detected in a screening test for PDAC.*

Response: The reviewer makes an excellent point. To address this concern, we repeated our ANN analysis using a revised training set. In this analysis, the six stage 1 samples used in the training set were exchanged with six late-stage PDAC samples consisting of five stage 3 and stage 4 samples and one stage 2B sample. This maintained the sample size for the PDAC group in the revised training set and allowed us to construct a revised test set that included only stage 1 (n=7) and stage 2 (n=7) PDAC samples. The training of the ANN was repeated in its entirety so that the results are comparable to the original ANN analysis. The test samples were not exposed to the ANN during the training process, although these were no longer ‘blinded’ as we knew the sample identity.

Results of the analysis are presented in Supplementary Figure 5A. The classification accuracy of the PDAC samples in the test set that now only had stage 1 and 2 samples improved from 87.5% to 100%, with no misclassification. Replacing the early stage samples in the original training set with late stage samples appeared to have increased the accuracy of the ANN approach possibly due to the larger number of spectral pattern differences occurring in the late-stage samples that encompassed spectral patterns in early-stage spectra.

To understand the effect of inclusion versus exclusion of early-stage samples on the salient spectra that contribute to the PDAC classification, we repeated the salient spectral region analysis. The results are presented in Supplementary Figure 5B as a comparison against the original salient spectral region results (Figure 4A). These results indicate that spectral patterns in the early-stage spectra may be effectively represented within the larger number of spectral pattern changes associated with the late-stage PDAC spectra. [Please see pages 13-14, lines 530 -559.](#)

• *Please change reference 5 to a more recent paper.*

Response: We have changed reference 5 to a more recent paper.

Reviewer #2 (Remarks to the Author):

Solaiyappan et al. presented an excellent work utilizing a neural network for PDAC Dx of 1H MR spectral patterns of plasma metabolites. The study included PDAC, benign disease, and controls and conducted a multi-class classification, an adequate design for translational biomarker research.

Response: We thank the reviewer for these comments.

A few comments are listed below for further consideration:

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Response: We agree with the reviewer. We have included and discussed reviews in the revised manuscript. We have removed the discussion related to proteomics. [Please see pages 3-4, lines 88 – 117.](#)

- Studies using spectroscopy usually aim to extract meaningful signals with assigned metabolites. This study, instead, focused on patterns. I guess it would be nice to discuss the pros/cons/trade-offs.

Response: The reviewer makes an excellent point. We have revised the manuscript to include the pros/cons/trade-offs. [Please see page 12, lines 479 - 509.](#)

- (Line 425-436) How are we certain that the neural-net selected the patterns in meaningful areas? Can we do anything to at least provide or speculate on the nature (e.g., possible classes of compounds) of the selected regions? Can we know which regions are of more importance than the rest and where misclassification may occur?

Response: Spectral regions that played a role in discriminating between malignant, disease and normal groups were associated with lipids, glucose, lactate, acetate, glutamine, myoinositol, BHB and BCAA regions in both ZGPR and CPMG spectra. Macromolecular regions were clearly identified as contributing to classification accuracy, but it was difficult to identify the specific macromolecules within these regions. Furthermore, the broad spectral pattern of the macromolecules may alter the overlapping spectral features of those metabolites that may appear in the same region. This can interfere with estimating concentrations of these metabolites to confirm their role in classification. Overlapping metabolites that we were able to identify in these macromolecular regions (Figure 4) included: tyrosine, phenylalanine, histidine, threonine, BHB, proteins and lipids. In addition to macromolecules, signals from glucose may also appear at different spectral ranges that overlap with other metabolites and contribute to the classification. Metabolites that overlap with glucose-related spectral regions included glycine and taurine. We found that glucose clusters I & II played an important role in the discrimination. We have included these points in the revised manuscript. [Please see pages 12, lines 464 - 477.](#)

- *The raw data and the analytical scripts should be publicly available, as it is generally a requirement for this category of studies.*

Response: We will make the code and data publicly available, with relevant academic licensing permission, before publication. We have provided this statement in the revised manuscript. [Please see page 14, lines 574 - 578.](#)